A3 Project

Project Name: Strengthen IO
Purpose: This program is designed to aid people in building new skills in different areas and fields.
How it works: Someone can prompt the AI to create a course for skills across various fields. These courses can be structured in different ways, such as a typical online course with lessons, a course that helps you build a project while learning, and others tailored to your needs. The AI's name is "Billy".
Examples:

- I want to learn how to create a website.
- Teach me to do my own taxes.
- Walk me through building a project for RAG.

Features:

- Users can give sources and have the AI build a course around those sources.
- Users can specify the tone, pace, and length of the course, as well as specify the best way they can learn (primarily auditory, visual, or text-based).
- Assignments are offered during the course to apply skills as you learn them.
    - Users can modify the difficulty of each assignment for more relaxing or challenging questions.
- An AI assistant is built into each course for clarification at any point.
    - The chats are saved to each course as well for reference purposes.
- Users have the option to create a course tailored around building a specific project or fictitious scenario.
- During the course, if a user feels they did not fully understand the lesson/idea/topic, they have the option to regenerate that lesson or create more assignments.
- Users can request skills for specific careers or job descriptions.
- Built-in notebook for each course

Technology:

- Web-app
- Gemini API
- Several AI Agents
    - Researcher
    - Judge
    - Builder
    - Orchestrator

Step-by-step Walkthrough:

- User searches the site and opens up to a landing page

- They click log in/create account in the top right corner
    - Creating an account will prompt a series of questions beyond just email, phone number, and general information
        - What topics interest you the most? (options such as tech, history, finances, etc.)
        - What kind of a learner are you? (options such as fast-learner, slow and steady, etc.)
        - What tools help you learn best? (options such as AI assistance, note-taking, videos, reading, audio recordings, etc.)
- After logging in, users have a dashboard that shows their current courses, completed courses, and a button to create a new course
- When clicking "Create a new course", they set the options for the course
    - Percentage of videos, reading, audio recordings, and visuals
    - How much history they want, such as background on the material
    - The tone, pace, and length of the course
    - Amount (a little, a bit, a lot, etc.) and difficulty of assignments
    - Previous experience on the topic that they have
    - The prompt for the actual course
    - Whether they want to base the course on specific sources, have the AI decide how to teach them, or both
- They click "Create," and the course gets generated (during this process, there should be text on the screen shuffling phrases such as "adding some touch-ups," "give me a minute," etc.)
    - The orchestrator agent is to handle all communication between the AI agents
    - This starts with the researcher agent receiving all of the user's options and prompts. Then, it either uses the provided sources to build the course or looks across the internet for materials, sites, and documentation.
    - Once it finds all of this, the information is sent back to the orchestrator, which sends it to the judge for review to see whether it matches the prompt and is adequate information.
        - If it is not up to par, the judge sends a response to the orchestrator, which relays it to the researcher.
    - Once the judge approves, the information is sent to the "builder" for course creation.
    - Finally, the information gets relayed back to the orchestrator, who turns the builder's response into code.
- A preview of the course shows up, and they can modify it before solidifying the course
    - The course is structured as modules and lessons within those modules. The preview allows the user to click through some of the modules and lessons to see how they look.

        - This same screen should also have the AI chatbot agent to help with tweaks.

Theme:

- Colors:
    - Charcoal Grey
    - Navy Blue
    - Olive Green
- Primarily picture based, less heavy on emojis
- Minimalist design